LIKHA cooldown 1b: technical report (2026-11-06)

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Status: finished, fetched, sha-verified, evaluated. INTERNAL, no release, no served endpoint. Plan reference: `docs/LIKHA-STAGE2-PLAN-2026-11-06.md`, Option 1. Evaluation details: `experiments/bio-behavior-2026-11-05/LIKHA-STAGE1-PROBES.md`.

1 What was run

Question: what was the missing learning-rate anneal of stage 1 worth? Method: load the stage-1 weights (fp32 export sha256 47fc4277…), reset the optimizer (fresh AdamW), warm up 200 steps to lr 4e-5 (the stage-1 final lr), cosine to 4e-6 over exactly 20,000 steps of 32,768 tokens = 655,360,000 tokens, one pass over a uniform random 40% subsample of every corpus shard (same corpus, sha ab5bfe7e…; the train/val split is unchanged: seed 1337, 0.5%, so val sequences are never trained on), natural-proportion mix, tag dropout 0.1, bf16, 1x H100 80GB SECURE, sdpa attention, seed 1337.

Code change (devops): `train.py` gained `--init-weights` / `--init-weights-sha` (sha256 verified, no pickle, fresh optimizer) and `--subsample`; `data.py::mixed_records` gained a deterministic uniform subsample. Selftest 57 checks OK; CPU smoke test loaded the weights (loss 1.4 to 1.7 at step 1 to 2). Bundle sha256 eda615a2b038e426…; launcher and pod script in `scripts/fleet/likha_cooldown/`. The subsample replaced the file-order prefix, which would have given a biased 40% of each shard.

2 Outcome

Stop reason OK (rc 0) at step 20,000. Pod terminated (404), spend USD 11.09 (166.8 min, ledger row), under the USD 15 cap and below the USD 13 estimate. Throughput 70,000 tokens/s, peak VRAM 31.6 GB, no non-finite or clipped gradients (grad norm 0.3 to 0.5).

Step	val_loss	ppl agri	ppl industrial (839 val tokens, noise)
stage 1 final	1.5344	4.59	11.09
2,000	1.5459	4.64	10.54
4,000	1.5170	4.52	8.37
8,000	1.5079	4.49	6.59
10,000	1.5478	4.67	8.15 (transient bump, recovered)
14,000	1.4896	4.41	6.92
18,000	1.4844	4.39	6.99
20,000 (final)	1.4829	4.38	7.09

3 Did it help? (stage 1 vs cooldown, same 5,234 val sequences, paired)

The val set is a stratified sample of the stage-1 VAL split (5,000 random plant sequences plus all 234 industrial sequences of 50 to 500 residues), binned by 8-mer containment against all TRAIN records (a homology proxy; MMseqs2 clusters do not exist yet). Mean per-sequence NLL change, cooldown minus stage 1, with bootstrap 95% CI:

Bin	n	change (nats/residue)	95% CI
plant, novel (containment < 0.1)	1,230	−0.013	−0.015 to −0.012
plant 0.1 to 0.3	1,382	−0.041	
plant 0.3 to 0.6	1,585	−0.054	
plant near-copy (>= 0.6)	803	−0.064	−0.070 to −0.058
industrial, novel	49	+0.008	−0.008 to +0.028
industrial near-copy (>= 0.6)	97	−0.097	−0.130 to −0.066

Perplexity on novel plant sequences 13.42 to 13.24. Reading: the improvement is almost entirely on sequences that resemble the training data; on novel sequences it is 1.3% in perplexity. The anneal was not the main limit.

Memorisation went UP: nll of the 14 rice (trained) probes 0.634 to 0.541; greedy continuation of lysozyme (a corpus member) 0.11 to 0.97 identity; cooldown samples contain a 95% near-copy of a training sequence (stage 1: best 80%) and 4 of 60 samples at or above 50% identity against 1 of 60 for stage 1. (One protein of nine did the opposite, Q6K9A2 0.93 to 0.50; n is small.)

Folding (ESMFold v1, audited commit, controls in the same run; details in the probes file): confident folds (mean pLDDT >= 70) 3 of 60 samples for stage 1 and 5 of 60 for cooldown, but 3 of the 5 cooldown ones are near-copies or homologs of training sequences; novel and confidently folded is 2 of 60 for both models. No real difference in generating novel foldable sequences.

4 Decision and caveats

Recommendation: do NOT adopt the cooldown as an upgrade; keep the stage-1 export as the base. The cooldown increased memorisation for a 1.3% perplexity gain on novel sequences. The lesson for stage 2 is data diversity and deduplication, not more epochs on the same data. Caveats: 8-mer containment is a proxy; industrial novel n is 49; probe sets are small and chosen by the author; one run, no seeds repeated; sampling at T=1.0 without top-k is the harshest setting.

5 Artifacts (verified)

`results.tgz` sha256 `67b3b765478cf342a78fa6059f4f1c37647743a7d06b83685eb3cf354b12e5a9` equals the pod's. Export MANIFEST shas match after extraction: model.bf16 `6237232b3d29da59b9e6f231ffa25f412690021d04c558e6f8a49188e9d6e080`, model.fp32 `903d5db49486745774728a1397f9e6dd077846660e6cd65e8026f3bcacbbe925`, config.json (same as stage 1) `f613b425...`, tokenizer.json (same) `e17efa81...`, EXPORT.json `51af4ca8...`. Location `D:\PinoAI-data\likha\cooldown-1b-step20000\out\`. Folding run (separate job, USD 0.18, 12 min on a 4090): results.tgz sha256 `423f9e10133f2a8569be34dc28514ee4bc4118c74818c594af0a36adaa431f83`, location

`D:\PinoAI-data\likha\fold-1\out\` (jsonl + 165 PDBs); the jsonl is also in `experiments/bio-behavior-2026-11-05/results/likha_fold_results.jsonl`.

Spend this week after both jobs: USD 190.02 of USD 300 (runner sum, duplicates excluded).